PROJECT REPORT

Team No: 16

Names and Roll no:

N. Priyadarshini (CS22B1009)

G. Satya Priya (CS22B1012)

K. Sri Harsha Priya (CS22B1017)

## Topic: Optimizing Branch Prediction Technique

Control Hazards impact the performance of a processor. The disruptions in instruction flow caused by branches or jumps are control hazards. And this leads to the Stalls and affect the performance.

To overcome the stalls, we will use Branch Predictions. These reduces the stalls and improves the instruction throughput in modern process.

## Optimizing Technique:

Here we proposed a new technique using 2*1 Multiplexer. The inputs of the multiplexer are A and B, selective input is S.

A – Output from 2-bit Branch prediction

B – Output from Static Branch prediction

S – Output from the Correlating Brach Prediction

Truth Table of the 2*1 Multiplexer:

| S | A | B | Out |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

## Procedure:

Here we have inputs S, A, B of the Multiplexer where S (the selective input) which is computed from correlating branch prediction.

In this case we will directly send the predicted value predicted by correlating branch prediction.

Coming to the inputs A, B these are derived from computing 2-bit and static but here instead of sending direct values we will compare them with the branch history. If the prediction is correct, we send 1 as input to the Multiplexer if not we send 0 as input. Then further the outcome of the multiplexer is calculated and compared with the branch history. 1 as taken and 0 as not taken. And mispredictions are calculated.

Here is the Example how our technique works.

## Example:

Branch Pattern History:

T N N T T N T N T T N

## Let's Calculate the outcome of B using Static branch prediction:

Let's assume its always taken, hence it will be T.

Branch history: T N N T T N T N T T N

Branch prediction: T T T T T T T T T T T

OUT COME : 1 0 0 1 1 0 1 0 1 1 0

(Correct is predicted as 1 and incorrect is predicted as 0)

## Calculate the outcome of A (2-bit prediction):

Branch history: T N N T T N T N T T N

Branch prediction: T T T NT T T T T T T

Outcome: 1 0 0 0 1 0 1 0 1 1 0

(Correct is predicted as 1 and incorrect is predicted as 0)

## Calculate the outcome of S (Correlating):

Branch history: T N N T T N T N T T N

Branch prediction: N N N N N N N N N N N

Outcome: 0 0 0 0 0 0 0 0 0 0 0

(for the correlating we will take direct prediction as 1 and not take as 0)

## OUTCOME OF OPTIMIZED TECHNIQUE:

Branch history: T N N T T N T N T T N

S : 0 0 0 0 0 0 0 0 0 0 0

A : 1 0 0 0 1 0 1 0 1 1 0

B : 1 0 0 1 1 0 1 0 1 1 0

R: 1 0 0 0 1 0 1 0 1 1 0

Here R is the result.

Predicted: T N N N T N T N T T N

Comparing the predicted values with Branch history we will get 1 misprediction using the optimized solution.

## COUNT OF MISPREDICTIONS:

- 1-BIT BRANCH PREDICTION : 7
- 2-BIT BRANCH PREDICTION : 6
- CORRELATING BRANCH PREDICTION : 6
- STATIC BRANCH PREDICTION : 5
- OPTIMIZED BRANCH PREDICTION : 1

## Conclusion:

Through the MUX, our technique dynamically chooses the most accurate prediction enhancing overall prediction accuracy and efficiency.